

Project on Freestyle Jobs

To Begin with the Lab

Summary of the Lab

This lab demonstrates deploying a Java web application using **Jenkins Freestyle Jobs** with two EC2 instances: a **Jenkins server** and a **Tomcat server**. The Tomcat server is configured by installing **Java (Amazon Corretto 1.8)** and **Apache Tomcat**, and verifying it through port **8080**. On the Jenkins server, the **Publish Over SSH plugin** and **Maven** are configured. A Freestyle job is created that pulls source code from **GitHub**, builds the project using **Maven (compile and package)**, and generates a WAR file. Jenkins then transfers the WAR file to the Tomcat server using SSH, deploys it in the **webapps** directory, and the application is accessed via the Tomcat URL.

- EC2 Instance (Two servers) are used:
- Jenkins Server:
 - Maven, Publish Over SSH plugin
- Tomcat Server:
 - Java, Apache Tomcat
- Create a new EC2 instance.
- Recommended configuration:
- Instance type:
 - t2.medium
- Storage:
 - 15 GB
- Security Group:
 - Allow HTTP (80),
 - Allow HTTPS (443),
 - Allow 8080, Allow SSH (22)
- Launch the instance.
- Connect to the server using SSH.
- [Install Java:](#)
 - `sudo yum install java-1.8.0-amazon-corretto-devel`

```
[ec2-user@ip-172-31-23-184 ~]$ sudo yum install java-1.8.0-amazon-corretto-devel
Amazon Linux 2023 Kernel Livepatch repository                               245 kB/s | 30 kB   00:00
Dependencies resolved.
-----
Package                                Architecture    Version                               Repository        Size
-----
Installing:
java-1.8.0-amazon-corretto-devel        x86_64          1:1.8.0_482.b08-1.amzn2023          amazonlinux        63 M
Installing dependencies:
adwaita-cursor-theme                    noarch          47.0-1.amzn2023.0.1                 amazonlinux        325 k
```

- Verify:
- `java -version`

```
[ec2-user@ip-172-31-23-184 ~]$ java -version
openjdk version "1.8.0_482"
OpenJDK Runtime Environment Corretto-8.482.08.1 (build 1.8.0_482-b08)
OpenJDK 64-Bit Server VM Corretto-8.482.08.1 (build 25.482-b08, mixed mode)
[ec2-user@ip-172-31-23-184 ~]$
```

- Install Apache Tomcat

- [Download Tomcat:](#)

Binary Distributions

- Core:
 - [zip \(pgp, sha512\)](#)
 - [tar.gz \(pgp, sha512\)](#)
 - [32-bit Windows zip \(pgp, sha512\)](#)
- `wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.115/bin/apache-tomcat-9.0.115.tar.gz`
- Extract the package:
 - `tar xvzf apache-tomcat-9.0.115.tar.gz`
- Move inside Tomcat directory:
 - `cd cd apache-tomcat-9.0.115/bin/`
- Look for the startup.sh file

```
[ec2-user@ip-172-31-23-184 ~]$ cd apache-tomcat-9.0.115/bin/
[ec2-user@ip-172-31-23-184 bin]$ ls
bootstrap.jar  catalina.sh  commons-daemon-native.tar.gz  configtest.sh  digest.sh  setclasspath.bat  shutdown.sh  tomcat-juli.jar  tool-wrapper.sh
catalina-tasks.xml  ciphers.bat  commons-daemon.jar  daemon.sh  makebase.bat  setclasspath.sh  startup.bat  tomcat-native.tar.gz  version.bat
catalina.bat  ciphers.sh  configtest.bat  digest.bat  makebase.sh  shutdown.bat  startup.sh  tool-wrapper.bat  version.sh
[ec2-user@ip-172-31-23-184 bin]$
```

- Start Tomcat:

- `./startup.sh`

```
[ec2-user@ip-172-31-23-184 bin]$ ./startup.sh
Using CATALINA_BASE:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_HOME:   /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_TMPDIR: /home/ec2-user/apache-tomcat-9.0.115/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/ec2-user/apache-tomcat-9.0.115/bin/bootstrap.jar:/home/ec2-user/apache-tomcat-9.0.115/bin/tomcat-juli.jar
Tomcat started.
[ec2-user@ip-172-31-23-184 bin]$
```

- Check Tomcat in browser:

- `http://<TOMCAT_PUBLIC_IP>:8080`

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/9.0.115

If you're seeing this, you've successfully installed Tomcat. Congratulations!

Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Developer Quick Start

- Tomcat Setup
- Realms & AAA
- Examples
- Servlet Specifications
- First Web Application
- JDBC DataSources
- Tomcat Versions

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the manager application is split between different users. [Read more...](#)

[Release Notes](#)

[Changelog](#)

[Migration Guide](#)

Documentation

- [Tomcat 9.0 Documentation](#)
- [Tomcat 9.0 Configuration](#)
- [Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

- [Tomcat 9.0 Bug Database](#)
- [Tomcat 9.0 IssueTracker](#)

Getting Help

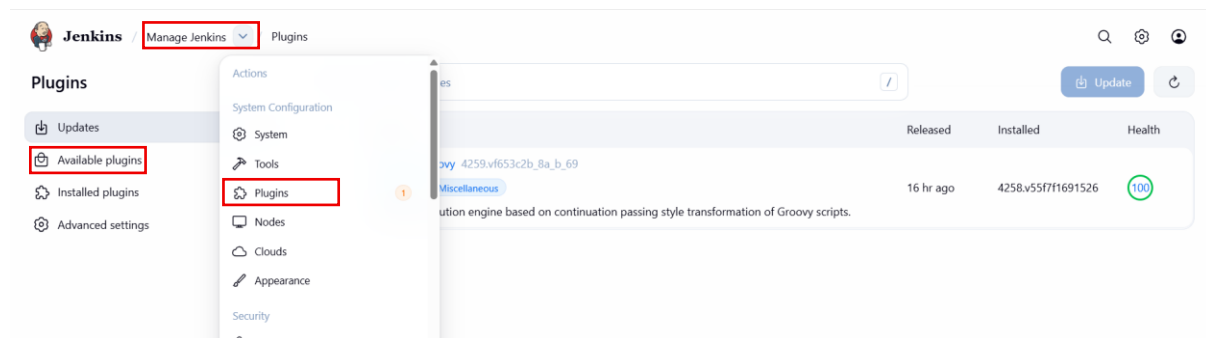
[FAQ and Mailing Lists](#)

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [taglibs-user](#)
User support and discussion for [Apache Taglibs](#)
- [tomcat-dev](#)
Development mailing list, including commit messages

34230.36.978080/docs/cluster-howto.html

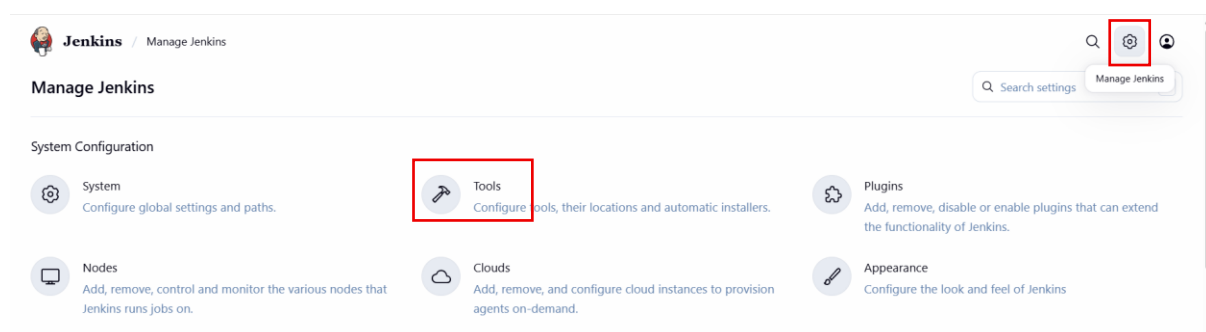
- If Tomcat page appears → installation successful.
- Stop Tomcat (for deployment):
 - ./shutdown.sh
- Install Jenkins Plugin (Publish Over SSH)
- Go to Jenkins Dashboard
- Manage Jenkins
- Plugins
- Available Plugins



- Search: Publish Over SSH



- Install the plugin.
- Open **Jenkins Dashboard**.
- Click **Manage Jenkins**.
- Click **Tool**.



- Scroll to **Maven Installation**.
- Click **Add Maven**.

Maven installations

[+ Add Maven](#)

Maven

Name

Maven

☒ Install automatically ?

Install from Apache

Version

3.9.13

[+ Add Installer](#)

- Click Save.
- If you have already done with part you can skip.
- Go to Jenkins Dashboard.
- Click New Item.

New Item

Enter an item name

Maven-Freestyle

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

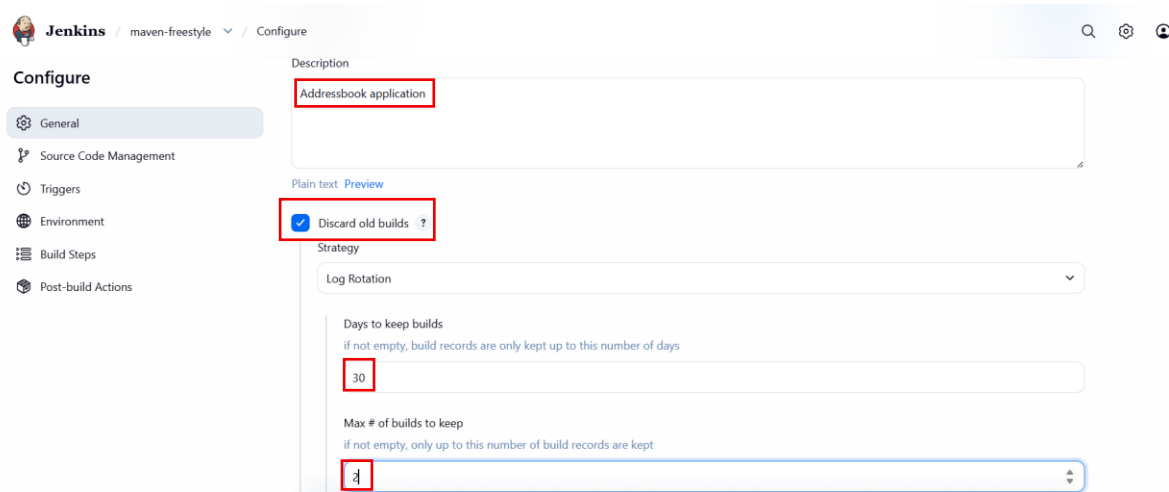
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

OK



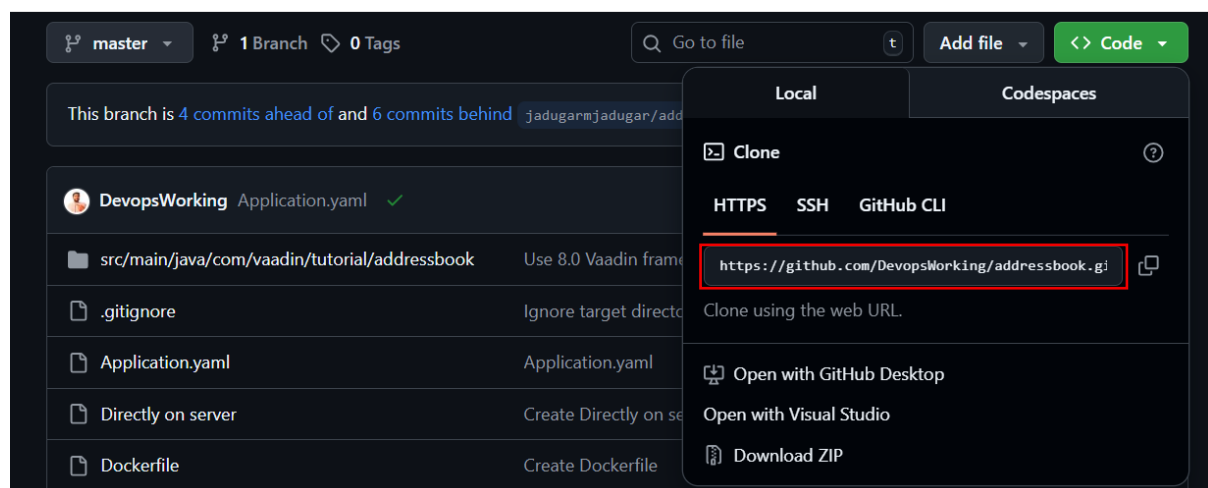
- Navigate Source Code Management
- Select the git option from there

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

- ☐ None
☒ Git

- After selecting git.
- Copy the URL from the github



- Enter the Repository URL
- Note: If your repository is private, you have to select the credentials from the credential option.

Repositories ?

Repository URL ?

`https://github.com/DevopsWorking/addressbook.git`

❗ Please enter Git repository.

Credentials ?

- none -

+ Add

Advanced ▾

- Check you github repository is master or main.

Branches to build ?

Branch Specifier (blank for 'any') ?

`*/master`

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☒ Delete workspace before build starts

Advanced ▾

- ☐ Use secret text(s) or file(s) ?
- ☐ Send files or execute commands over SSH before the build starts ?
- ☐ Send files or execute commands over SSH after the build runs ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans

- Navigate to the Build Step section
- Click on the Add Build Step
- Select the 'Invoke top-level Maven targets'

Environment

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

+ Add build step

Filter

Execute Windows batch command

Execute shell

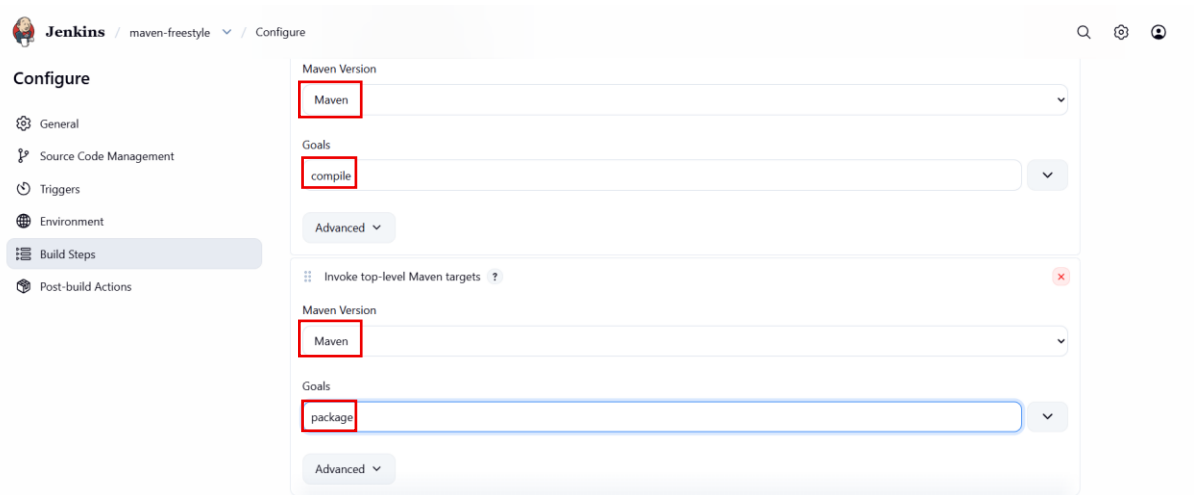
Invoke Ant

Invoke Gradle script

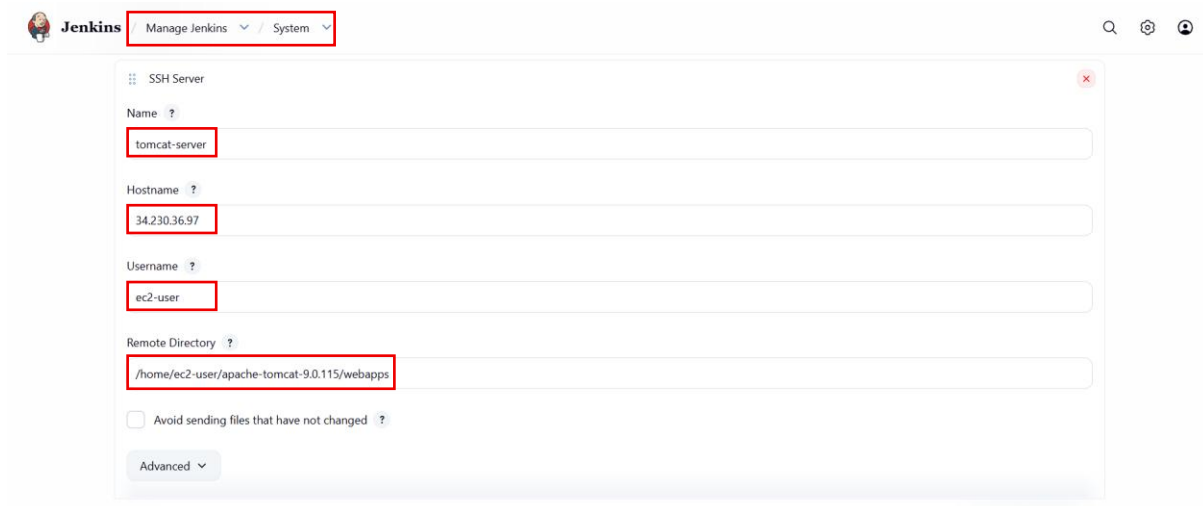
Invoke top-level Maven targets

Run with timeout

Set build status to "pending" on GitHub commit



- Similarly make Goals for – compile and package
- Save the Job
- Now click on the Build now
- After saving it will create a job successfully
- Configure Publish Over SSH
- Go to:
 - Manage Jenkins
- Configure System
- Scroll to:
 - Publish Over SSH
- Click Add Server.
- Fill details:
- Name:
 - TomcatServer
- Hostname:
 - <Tomcat_Public_IP>
- Username:
 - ec2-user
- Remote Directory:
 - /home/ec2-user/apache-tomcat/webapps



Jenkins Manage Jenkins System

SSH Server

Name ?
tomcat-server

Hostname ?
34.230.36.97

Username ?
ec2-user

Remote Directory ?
/home/ec2-user/apache-tomcat-9.0.115/webapps

☐ Avoid sending files that have not changed ?

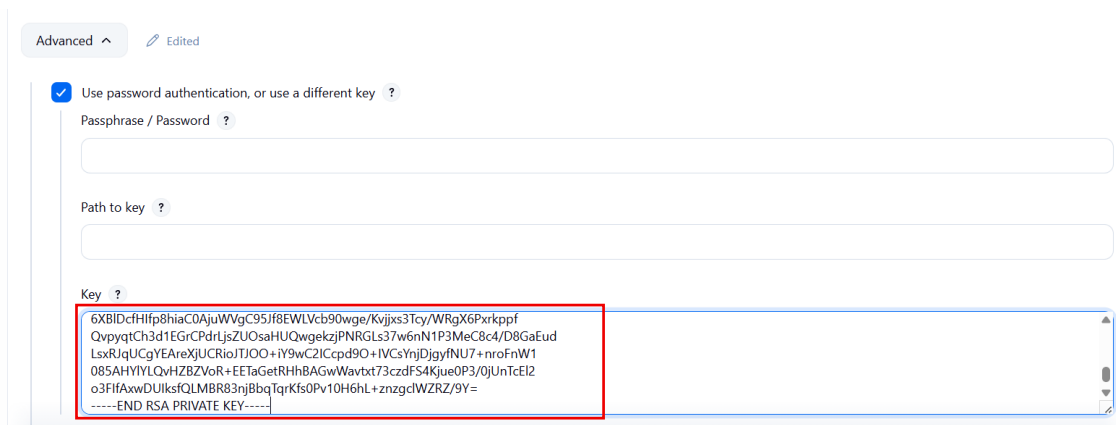
Advanced

```

[ec2-user@ip-172-31-23-184 bin]$ cd ..
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd ..
[ec2-user@ip-172-31-23-184 ~]$ cd apache-tomcat-9.0.115/
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd w
webapps/ work/
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd webapps/
[ec2-user@ip-172-31-23-184 webapps]$ pwd
/home/ec2-user/apache-tomcat-9.0.115/webapps
[ec2-user@ip-172-31-23-184 webapps]$

```

- In Advanced section:
 - Use password authentication or use a different key
- Add Private Key
- Paste the EC2 private key (.pem file).



Advanced Edited

☒ Use password authentication, or use a different key ?

Passphrase / Password ?

Path to key ?

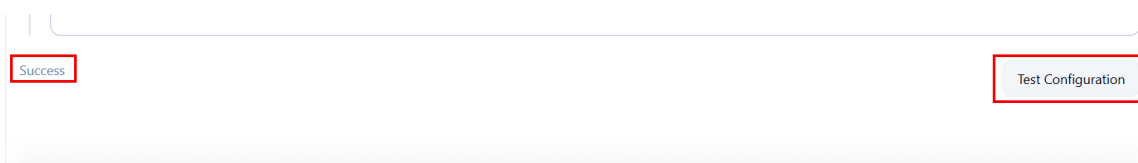
Key ?

```

6XBIDcfHfp8hiaC0AjuWVgC95Jf8EWLvc90wge/Kvijxs3Tcy/WRgX6Pxrppf
QvpyqtCh3d1EGrCPdrljsZUOsaHUQwgekzjPNRGLs37w6nN1P3MeC8c4/D8GaEud
LsxRJqUCgYEAReXjUCRioITJOO+iY9wC2ICpd9O+IVCsYnjDjgyfNU7+nroFnW1
085AHYIYLQvHZBZVoR+EEtaGetRHhBAGwWavtxt73czdFS4Kjue0P3/0jUnTcEI2
o3FIHxwDUlksQLMBR83njBbqTqrKfs0Pv10H6hL+znzgcIWZRZ/9Y=
-----END RSA PRIVATE KEY-----

```

- Click:
 - Test Configuration
- If configuration is correct → Success message appears.



Success

Test Configuration

- Click Save.
- Open Jenkins job > Click Configure

- Configure Artifact Transfer
- Look for **Post-build Actions**
- Select:
 - Send build artifacts over SSH
- Choose server:
 - Tomcat-Server
- Provide source file:
 - target/addressbook-2.0.war
- Remove prefix:
 - target/

Send build artifacts over SSH ?

SSH Publishers

SSH Server

Name ?

tomcat-server

Advanced

Transfers

Transfer Set

Source files ?

addressbook-2.0.war

Remove prefix ?

target/

- Click Save.
- Click on Build Now
- In console Output shows:
 - Transferred 1 file

```
SSH: Connecting from host [ip-172-31-13-72.ec2.internal]
SSH: Connecting with configuration [tomcat-server] ...
SSH: Disconnecting configuration [tomcat-server] ...
SSH: Transferred 1 file(s)
Finished: SUCCESS
```

- Verify WAR File on Tomcat Server
- Login to Tomcat server:
 - ls
- You should see:
 - addressbook-2.0.war

```
[ec2-user@ip-172-31-23-184 webapps]$ ls
ROOT addressbook-2.0.war docs examples host-manager manager
[ec2-user@ip-172-31-23-184 webapps]$ ls -ll
total 13836
drwxr-xr-x. 3 ec2-user ec2-user 16384 Mar 12 14:52 ROOT
-rw-rw-r--. 1 ec2-user ec2-user 14133230 Mar 12 16:48 addressbook-2.0.war
drwxr-xr-x. 16 ec2-user ec2-user 16384 Mar 12 14:52 docs
drwxr-xr-x. 7 ec2-user ec2-user 99 Mar 12 14:52 examples
drwxr-xr-x. 6 ec2-user ec2-user 79 Mar 12 14:52 host-manager
drwxr-xr-x. 6 ec2-user ec2-user 114 Mar 12 14:52 manager
[ec2-user@ip-172-31-23-184 webapps]$ cd bin/
-bash: cd: bin/: No such file or directory
[ec2-user@ip-172-31-23-184 webapps]$ cd ..
[ec2-user@ip-172-31-23-184 apache-tomcat-9.0.115]$ cd bin/
[ec2-user@ip-172-31-23-184 bin]$ ./startup.sh
Using CATALINA_BASE: /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_HOME: /home/ec2-user/apache-tomcat-9.0.115
Using CATALINA_TMPDIR: /home/ec2-user/apache-tomcat-9.0.115/temp
Using JRE_HOME: /usr
Using CLASSPATH: /home/ec2-user/apache-tomcat-9.0.115/bin/bootstrap.jar:/home/ec2-user/apache-tomcat-9.0.115/bin/tomcat-juli.jar
Tomcat started.
[ec2-user@ip-172-31-23-184 bin]$
```

- Access Application
- Open browser:
 - http://<Tomcat_Public_IP>:8080/addressbook-2.0
- The Address Book Application UI should appear.

First Name	Last Name	Email
George	White	george@white.com
Daniel	Thompson	daniel@thompson.com
Timothy	Jones	timothy@jones.com
Peter	Wilson	peter@wilson.com
Dan	Robinson	dan@robinson.com
Dan	Davis	dan@davis.com
Olivia	Davis	olivia@davis.com
Dan	Smith	dan@smith.com
Daniel	Anderson	daniel@anderson.com
Alice	Thomas	alice@thomas.com
Linda	Harris	linda@harris.com
Daniel	Robinson	daniel@robinson.com
Mike	Young	mike@young.com
Umberto	Anderson	umberto@anderson.com
Scott	Thompson	scott@thompson.com
Rene	Martin	rene@martin.com
Lisa	Martin	lisa@martin.com

